

Number Theory

CS1010 Discrete Mathematics for Computer Science

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Modular Arithmetic: Exercise

2. Show that if a, b are positive integers, then

$$(2^a - 1) \bmod (2^b - 1) = 2^{a \bmod b} - 1.$$

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A. Write $a = qb + r$, where $0 \leq r < b$. Then

$$2^a - 1 = 2^{qb+r} - 1 = 2^r(2^b)^q - 1 \equiv 2^r - 1 \pmod{2^b - 1}.$$

Hence, $(2^a - 1) \bmod (2^b - 1) = 2^r - 1$, where $r = a \bmod b$.

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Q. Show that $2^{35} - 1$ and $2^{34} - 1$ are relatively prime.

A. $\gcd(2^{35} - 1, 2^{34} - 1) = 2^{\gcd(35,34)} - 1 = 2^1 - 1 = 1$.



Uniqueness of Prime Factorization

Every integer greater than 1 can be written uniquely as a prime or as the product of two or more primes, where the prime factors are written in order of nondecreasing size.

Proof (by contradiction): Suppose positive integer n can be split into primes in two distinct ways:

$$n = p_1 p_2 \dots p_s = q_1 q_2 \dots q_t$$

Cancel all common primes until no primes are shared:

$$p_{i_1} p_{i_2} \dots p_{i_u} = q_{j_1} q_{j_2} \dots q_{j_v}$$

By Lemma 3, prime p_{i_1} must divide q_{j_k} for some k , meaning they are the same prime since both are prime numbers. This contradicts the fact that all common primes were removed.



Solving Linear Congruences

Motivating Systems of Linear Congruences

Imagine a warehouse coordinator sorting a shipment of computer chips. The exact total x is unknown, but counting them in different batch sizes yields distinct leftovers:

- ▶ When arranged in groups of **5**, there are **3** chips left over.
- ▶ When arranged in groups of **7**, there are **2** chips left over.
- ▶ When arranged in groups of **11**, there are **6** chips left over.

This translates directly into a **system of linear congruences**:

$$\begin{cases} x \equiv 3 \pmod{5} \\ x \equiv 2 \pmod{7} \\ x \equiv 6 \pmod{11} \end{cases}$$



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Linear Congruences

Definition

A congruence of the form $ax \equiv b \pmod{m}$, where m is a positive integer, a, b are integers, and x is a variable, is called a **linear congruence**.

Example: How do you find x such that $5 \cdot x = 2 \pmod{7}$?

Definition

An integer $\bar{a}$ such that $\bar{a}a \equiv 1 \pmod{m}$ is called an **inverse of a modulo m** .

To solve $ax \equiv b \pmod{m}$, multiply both sides by $\bar{a}$ (if it exists).

Example: $3 \cdot 5 = 15 \equiv 1 \pmod{7} \implies x \equiv 6 \pmod{7}$



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Inverse of a modulo m

Theorem

If a and m are relatively prime integers and $m > 1$, then an inverse of a modulo m exists. Furthermore, this inverse is unique modulo m .

Proof: Since $\gcd(a, m) = 1$, by Bézout's Theorem there exist s, t such that:

$$sa + tm = 1 \implies sa + tm \equiv 1 \pmod{m}$$

Since $tm \equiv 0 \pmod{m}$, we get:

$$sa \equiv 1 \pmod{m}$$

Consequently, s is the modular inverse of a .



Find the inverse of 55 modulo 89

$$89 = 55 \cdot 1 + 34 \quad \Rightarrow \quad 34 = 89 - 55 \cdot 1$$

$$55 = 34 \cdot 1 + 21 \quad \Rightarrow \quad 21 = 55 - 34 \cdot 1$$

$$34 = 21 \cdot 1 + 13 \quad \Rightarrow \quad 13 = 34 - 21 \cdot 1$$

$$21 = 13 \cdot 1 + 8 \quad \Rightarrow \quad 8 = 21 - 13 \cdot 1$$

$$13 = 8 \cdot 1 + 5 \quad \Rightarrow \quad 5 = 13 - 8 \cdot 1$$

$$8 = 5 \cdot 1 + 3 \quad \Rightarrow \quad 3 = 8 - 5 \cdot 1$$

$$5 = 3 \cdot 1 + 2 \quad \Rightarrow \quad 2 = 5 - 3 \cdot 1$$

$$3 = 2 \cdot 1 + 1 \quad \Rightarrow \quad 1 = 3 - 2 \cdot 1$$

$$2 = 1 \cdot 2 + 0 \quad (\text{Stop})$$

$$1 = 3 - 2$$

$$= 3 - (5 - 3)$$

$$= 2 \cdot 3 - 5$$

$$= 2(8 - 5) - 5$$

$$= 2 \cdot 8 - 3 \cdot 5$$

$\vdots$

$$= 13 \cdot 55 - 21 \cdot 34$$

$$= 13 \cdot 55 - 21(89 - 55)$$

$$= 34 \cdot 55 + (-21) \cdot 89$$

Therefore,

$$55^{-1} \equiv \boxed{34} \pmod{89}.$$

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$$13 = 8 \cdot 1 + 5 \quad \Rightarrow \quad 5 = 13 - 8 \cdot 1$$

$$8 = 5 \cdot 1 + 3 \quad \Rightarrow \quad 3 = 8 - 5 \cdot 1$$

$$5 = 3 \cdot 1 + 2 \quad \Rightarrow \quad 2 = 5 - 3 \cdot 1$$

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Using Inverses to Solve Congruences

Solve the linear congruence $4x \equiv 5 \pmod{9}$.

Solution: Multiply both sides by the modular inverse of 4 mod 9, which is -2 :

$$-2 \cdot 4x \equiv -2 \cdot 5 \pmod{9}$$

$$-8x \equiv -10 \pmod{9}$$

Since $-8 \equiv 1 \pmod{9}$ and $-10 \equiv 8 \pmod{9}$, we get:

$$x \equiv 8 \pmod{9}$$

The solutions are integers $x \equiv 8 \pmod{9}$, namely 8, 17, 26, ... and $-1, -10, -19, \dots$



How to Solve a System of Linear Congruences?

Given the following system of linear congruences:

$$\begin{cases} x \equiv 3 \pmod{5} \\ x \equiv 2 \pmod{7} \\ x \equiv 6 \pmod{11} \end{cases}$$

Find x modulo $m = 5 \cdot 7 \cdot 11 = 385$.

Chinese Remainder Theorem (CRT)

Let $m_1, m_2, \dots, m_n$ be pairwise relatively prime positive integers greater than one and $a_1, a_2, \dots, a_n$ arbitrary integers. Then the system:

$$x \equiv a_i \pmod{m_i} \quad (\text{for } i = 1, 2, \dots, n)$$

has a unique solution modulo $m = m_1 m_2 \dots m_n$.

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The Chinese Remainder Theorem (CRT)

The System of Congruences

Given a system of n linear congruences:

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$\vdots$$

$$x \equiv a_n \pmod{m_n}$$

Where the moduli are pairwise relatively prime:

$$\gcd(m_i, m_j) = 1 \quad \text{for } i \neq j$$

The CRT Construction Steps

1. Compute: $M = m_1 \cdot m_2 \cdots m_n$
2. For each k , compute: $M_k = \frac{M}{m_k}$
3. For each k , find modular inverses y_k such that:

$$M_k y_k \equiv 1 \pmod{m_k}$$

4. Solution: $x = \sum_{k=1}^n a_k M_k y_k \pmod{M}$



The Chinese Remainder Theorem (CRT)

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Example of Solving System using CRT

System: $x \equiv 3 \pmod{5}$, $x \equiv 2 \pmod{7}$, $x \equiv 6 \pmod{11}$.

1. $M = 5 \cdot 7 \cdot 11 = 385$

2. $M_1 = 385/5 = 77$; $M_2 = 385/7 = 55$; $M_3 = 385/11 = 35$.

3. Inverses:

▷ $77 \cdot y_1 \equiv 1 \pmod{5} \implies 2 \cdot y_1 \equiv 1 \pmod{5} \implies y_1 = 3$

▷ $55 \cdot y_2 \equiv 1 \pmod{7} \implies 6 \cdot y_2 \equiv 1 \pmod{7} \implies y_2 = 6$

▷ $35 \cdot y_3 \equiv 1 \pmod{11} \implies 2 \cdot y_3 \equiv 1 \pmod{11} \implies y_3 = 6$

4. Combine:

$$\begin{aligned}x &= a_1 M_1 y_1 + a_2 M_2 y_2 + a_3 M_3 y_3 \\&= 3 \cdot 77 \cdot 3 + 2 \cdot 55 \cdot 6 + 6 \cdot 35 \cdot 6 \\&= 693 + 660 + 1260 = 2613 \equiv 303 \pmod{385}\end{aligned}$$

The smallest simultaneous positive solution is 303.



Fermat's Little Theorem

Theorem

If p is prime and x is an integer not divisible by p , then:

$$x^{p-1} \equiv 1 \pmod{p}$$

Furthermore, for every integer x we have: $x^p \equiv x \pmod{p}$.

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A. By Fermat's Little Theorem, $7^{10} \equiv 1 \pmod{11}$. Thus:

$$7^{222} = (7^{10})^{22} \cdot 7^2 \equiv (1)^{22} \cdot 49 \equiv 1 \cdot 5 = 5 \pmod{11}$$

Hence, $7^{222} \bmod 11 = 5$.



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- Q.* Show using Fermat's little theorem that, for every positive integer n , $42 \mid (n^7 - n)$.
- A.* Since $42 = 2 \cdot 3 \cdot 7$, it is enough to show $n^7 \equiv n \pmod{2}$, $n^7 \equiv n \pmod{3}$, $n^7 \equiv n \pmod{7}$.

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► If $2 \mid n$, done! Otherwise, FLT gives $n^{2-1} \equiv 1 \pmod{2}$, so $n^7 \equiv n \pmod{2}$.

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- ▶ If $2 \mid n$, done! Otherwise, FLT gives $n^{2-1} \equiv 1 \pmod{2}$, so $n^7 \equiv n \pmod{2}$.
- ▶ If $3 \mid n$, done! Otherwise, FLT gives $n^{3-1} \equiv 1 \pmod{3}$, so $n^6 \equiv 1 \pmod{3}$ and hence $n^7 \equiv n \pmod{3}$.

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- ▶ If $7 \mid n$, done! Otherwise, FLT gives $n^{7-1} \equiv 1 \pmod{7}$, so $n^7 \equiv n \pmod{7}$.

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- ▶ If $7 \mid n$, done! Otherwise, FLT gives $n^{7-1} \equiv 1 \pmod{7}$, so $n^7 \equiv n \pmod{7}$.

A. Thus $2, 3, 7 \mid (n^7 - n)$. Since they are pairwise coprime, $42 \mid (n^7 - n)$.



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$\mathcal{A}$. Since $\gcd(3, 10) = 1$ and $\phi(10) = 4$, Euler's theorem gives $3^4 \equiv 1 \pmod{10}$.
Since $27 = 4 \cdot 6 + 3$, we have

$$3^{27} = (3^4)^6 \cdot 3^3 \equiv 1^6 \cdot 27 \equiv 7 \pmod{10}.$$

$$3^{27} \bmod 10 = 7$$

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► Since $7 \equiv 1 \pmod{2}$, $N \equiv 1 \pmod{2}$.

► $7 \equiv 2 \pmod{5}$ and by FLT, $2^4 \equiv 1 \pmod{5}$. Hence we only need $7^{7^7} \bmod 4$.

Last Digit: Reducing the Exponent

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Since $7 \equiv -1 \pmod{4}$, and 7^7 is odd,

$$7^{7^7} \equiv (-1)^{7^7} \equiv -1 \equiv 3 \pmod{4}.$$

Last Digit: Reducing the Exponent

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Since $7 \equiv -1 \pmod{4}$, and 7^7 is odd,

$$7^{7^7} \equiv (-1)^{7^7} \equiv -1 \equiv 3 \pmod{4}.$$

Therefore,

$$N = 7^{7^{7^7}} \equiv 2^3 \pmod{5}.$$

Hence,

$$\boxed{N \equiv 3 \pmod{5}}.$$

So we have

$$N \equiv 1 \pmod{2}, \quad N \equiv 3 \pmod{5}.$$

Last Digit: Chinese Remainder Theorem

We have

$$N \equiv 1 \pmod{2}, \quad N \equiv 3 \pmod{5}.$$

The numbers congruent to 3 (mod 5) modulo 10 are

3, 8.

Among these, only 3 is odd. Therefore,

$$N \equiv 3 \pmod{10}.$$

Hence the last digit of $7^{7^{7^7}}$ is

3.



Thank You for your
attention!

Any questions?